Introduction to Probability & Statistics

Assignment 1, 2026/27

Instructions

Submit your answers to all questions marked **Hand-in**. You should upload your solutions to the VLE as a *single pdf file*. Marks will be awarded for clear, logical explanations, as well as for correctness of solutions.
Solutions to questions marked have been released at the same time as this assignment, in case you want to check your answers or need a hint.
You should also look at the other questions in preparation for your Week 3 seminar.

Starters

These questions should help you to gain confidence with the basics.

S1. A bag contains six balls numbered 1–6. Two balls are pulled out of the bag (without replacement), and the outcome is recorded as (i, j) if ball i is drawn first, and ball j drawn second. How many possible outcomes are there?
Now suppose that balls numbered 1–4 are white, and the other two are black. How many outcomes are there in which the following occur?

a. both balls are white;
b. both balls have the same colour;
c. at least one of the balls is white?

Answer

Since we are sampling without replacement there are $6 \times 5 = 30$ different possible outcomes: $\{(1,2),(1,3),(1,4),\ldots,(2,1),(2,3),\ldots,(6,5)\}$.

a. Here we need to choose a white ball followed by another white ball: $4 \times 3 = 12$;
b. As in part (a), the number of ways in which two black balls can be chosen is $2 \times 1 = 2$. Hence the total number of outcomes for which both balls have the same colour is $12 + 2 = 14$.
c. Here we need either both balls to be white (which we've already counted), or exactly one to be white. There are $4 \times 2 = 8$ ways for the first ball to be white and the second black, and another $2 \times 4 = 8$ ways for the balls to come out in the opposite order; so there are $16 + 12 = 28$ possible outcomes altogether. (Alternatively, we know from part (b) that there are only two outcomes where both balls are black, and since there are 30 possible outcomes in total, 28 of these must include at least one white ball.)

S2. Hand-in

A bag contains ten balls numbered 1–10. Three balls are pulled out of the bag (without replacement), and the outcome is recorded as (i, j, k) if ball i is drawn first, ball j second, and ball k third. By Theorem 1.5, there are $P_3^{10} = 10!/7! = 10 \times 9 \times 8 = 720$ possible outcomes in total. In how many of these outcomes do the following occur:

a. all three chosen balls are numbered 1–6?

b. at least one of the chosen balls show a number which is a multiple of 3?

Answer

a. We need to count the number of ordered lists of length three, which can be formed from the set of six balls numbered 1–6. By Theorem 1.5, this is $P_3^6 = 6!/3! = 120$. **[2 marks]**
b. There are three balls with numbers which are multiples of 3, so there are $P_3^7 = 7 \times 6 \times 5 = 210$ outcomes in which all of the balls show numbers which are *not* a multiple of 3. This means that there are $720 - 210 = 510$ outcomes in which at least one of the balls is a multiple of 3. **[3 marks]**

S3. Hand-in
A lecturer has to split 15 students into three groups of 5 to complete an assignment. How many ways are there for them to achieve this?

Answer

The lecturer has to choose 5 students to form the first group: there are $C_5^{15} = \binom{15}{5}$ ways to achieve this. They then need to choose 5 of the remaining 10 to form the second group: there are $C_5^{10} = \binom{10}{5}$ ways to do this. Once this is done, the remaining 5 form the final group (no further choices to be made!). **[3 marks]** However, this assumes that there is a "first group", etc. Since the order of the groups doesn't matter, we have over-counted by a factor of $3!$. So the required number is $C_5^{15}C_5^{10}/3! = 15!/(5!5!5!3!) = 126,126$. **[2 marks]**

S4. 30 sweets are to be divided amongst five children. In how many ways can this be done? What if every child must be given at least two sweets each?

Answer

There are $C_{5-1}^{30+5-1} = 46,376$ ways of dividing the sweets (allowing for the possibility of some children receiving none). If every child must be given two sweets, then if we do this first of all there are 20 sweets left to be divided: there are $C_{5-1}^{20+5-1} = 10,626$ ways of doing this.

S5. Consider two events A and B. Let C be the event that "A or B occurs, but not both". Express C in terms of A and B, using only the basic operations "union", "intersection", and "complement".

Answer

There are many possible answers. Here are three:

$$\begin{aligned} C &= (A \cup B) \cap (A \cap B)^c \\ &= (A \cup B) \cap (A^c \cup B^c) \\ &= (A \cap B^c) \cup (B \cap A^c). \end{aligned}$$

S6. Hand-in
Let E, F and G be three events. Find expressions for these events:

a. At least one of F and G occurs;
b. F and G occur, but not E;
c. F or G occurs, as well as E;
d. At most two of the events occur.

Answer

a. $F \cup G$; **[1 mark]**
b. $F \cap G \cap E^c$; **[1 mark]**
c. $E \cap (F \cup G)$; **[1 mark]**
d. This is the same as saying that "not all three events occur": $(E \cap F \cap G)^c$. **[2 marks]**

(You may have slightly different answers to the above, but if so then you should be able to prove that your versions are equivalent to mine!)

S7. Two dice are thrown. Let A be the event that the sum of the dice is even; let B be the event that the first die shows a higher number than the second; and let C be the event that the sum of the two dice is 6. List all the outcomes belonging to the events $A \cap B$, $A \cup C$, $A \cap C^c$, $A^c \cap C$, $A \cap B \cap C$.

Answer

Writing (i, j) for the outcome of the two dice, where i is the score on the first die and j the score on the second, we have:

$$\begin{aligned}
A \cap B &= \{(3,1),(4,2),(5,1),(5,3),(6,2),(6,4)\}; \\
A \cup C = A &= \{(1,1),(1,3),(1,5),(2,2),(2,4),(2,6),(3,1),(3,3),(3,5),(4,2),(4,4), \\
&\qquad (4,6),(5,1),(5,3),(5,5),(6,2),(6,4),(6,6)\}; \\
A \cap C^c &= \{(1,1),(1,3),(2,2),(2,6),(3,1),(3,5),(4,4),(4,6),(5,3),(5,5), \\
&\qquad (6,2),(6,4),(6,6)\}; \\
A^c \cap C &= \emptyset; \\
A \cap B \cap C &= \{(4,2),(5,1)\}.
\end{aligned}$$

Mains

These are a bit harder: you should aim to be able to do all of these by the end of the course.

M1. Hand-in

Consider drawing paths in the plane, starting from the point $(0, 0)$, such that each step either consists of going one unit up or one unit to the right.

a. How many paths are there from $(0, 0)$ to the point $(20, 20)$? (Hint: think of writing a path as a sequence $(U, R, R, ...)$ representing *Up* or *Right* steps.)
b. How many of the paths in part a. go through the point $(10, 10)$?

Answer

a. A path from $(0, 0)$ to $(20, 20)$ must contain exactly 20 Up steps and 20 Right steps. We just need to count how many ways there are to place the 20 Up steps amongst the 40 total steps: $\binom{40}{20} (= 137,846,528,820)$. **[2 marks]**
b. Arguing as in part a, there are $\binom{20}{10}$ paths from $(0, 0)$ to $(10, 10)$, and the same number from $(10, 10)$ to $(20, 20)$. By the basic counting principle, that gives a total of $\binom{20}{10}^2 (= 34,134,779,536)$ paths from $(0, 0)$ to $(20, 20)$ which pass through $(10, 10)$. **[3 marks]**

M2. A student has 1 bookshelf, 10 novels (among which 3 titles appear twice each), 4 textbooks on probability theory and 6 knitting books. In how many ways can he put the books on the shelf such that books in the same category (novels, textbooks, knitting books) are next to each other? Possibilities where the doubles are switched are considered to be the same and are not counted separately.

Answer

The 3 categories can be placed in $3! = 6$ ways. The 4 textbooks can be placed in $4! = 24$ ways, and the 6 knitting books in $6! = 720$. For the novels, by Proposition 1.4 there are $10!/(2!2!2!)$ ways of arranging them. This gives a total of $3!4!6!\frac{10!}{2!2!2!} = 47,029,248,000$ ways to order all books.

M3. Prove, by expanding in terms of factorials, that

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \qquad 1 \leq r \leq n\,.$$

Answer

$$\begin{aligned}\binom{n-1}{r-1} + \binom{n-1}{r} &= \frac{(n-1)!}{(n-r)!(r-1)!} + \frac{(n-1)!}{(n-1-r)!r!} \\ &= \frac{(n-1)!}{(n-r)!r!}(r + (n-r)) \\ &= \frac{n!}{(n-r)!r!} = \binom{n}{r}\,.\end{aligned}$$

M4. Consider a group of n objects, labelled $1, \ldots, n$. How many (unordered) sets of size r are there that contain object 1? How many are there that *don't* contain object 1? Use these observations to re-prove the identity in the previous question.

Answer

There are $\binom{n-1}{r-1}$ sets of size r that contain object 1. (First put object 1 in the set, and then choose another $r-1$ objects from the remaining $n-1$ objects available.) There are $\binom{n-1}{r}$ sets that don't contain object 1. There are a total of $\binom{n}{r}$ unordered sets of size r possible with n objects, and since all of these must either contain object 1 or not contain it, we see that

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \qquad 1 \leq r \leq n\,.$$

M5. 🛠 Prove that, for sets E and F:

a. $E \cap F \subseteq E \subseteq E \cup F$
b. if $E \subseteq F$ then $F^c \subseteq E^c$
c. $E = (E \cap F) \cup (E \cap F^c)$, and show that this union is disjoint.

Answer

a. If $\omega \in E \cap F$, then $\omega \in E$ (and $\omega \in F$), and so $E \cap F \subseteq E$. Similarly for $E \subseteq E \cup F$.
b. Let $\omega \in F^c$, and suppose that $\omega \in E$ also. Then, since $E \subseteq F$, we would have $\omega \in F$. But this would imply that $\omega \in F$ *and* $\omega \in F^c$, i.e. that $\omega \in F \cap F^c = \emptyset$, which is a contradiction. Thus $\omega \notin E$, i.e. $F^c \subseteq E^c$.
c. We can write
$$(E \cap F) \cup (E \cap F^c) = E \cap (F \cup F^c) = E \cap \Omega = E\,.$$
Furthermore, we know that $F \cap F^c = \emptyset$, and so $(E \cap F) \cap (E \cap F^c) = E \cap (F \cap F^c) = \emptyset$ (by part (a)). Thus the union is certainly disjoint.

M6. Prove the second of De Morgan's laws: for sets $E_1, \ldots, E_n$:

$$\left(\bigcap_{i=1}^{n} E_i\right)^c = \bigcup_{i=1}^{n} E_i^c \,.$$

Answer

As usual, we need to show that any element ω belonging to the first set belongs to the second, and vice versa.
Suppose first that $\omega \in (\bigcap_{i=1}^n E_i)^c$. Then ω does not belong to *all* of the sets E_i, and so there must be at least one set, say E_j, with $\omega \notin E_j$ (i.e. with $\omega \in E_j^c$). But this implies that $\omega \in \bigcup_{i=1}^n E_i'$.
Working in the other direction, suppose that ω belongs to the set on the RHS. Then there is at least one set E_k with $\omega \in E_k^c$. Thus $\omega \notin \bigcap_{i=1}^n E_i$, since

$$E_k^c \cap \left(\bigcap_{i=1}^{n} E_i\right) \subseteq (E_k^c \cap E_k) = \emptyset \,.$$

Therefore $\omega \in (\bigcap_{i=1}^n E_i)^c$, as required.

M7. Let $\mathbb{P}$ be a probability function and A and B events. Show that

$$\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A \cap B) \,.$$

Answer

$$\begin{aligned} \mathbb{P}(A^c \cap B^c) &= \mathbb{P}((A \cup B)^c) && \text{by De Morgan's law} \\ &= 1 - \mathbb{P}(A \cup B) && \text{by (P4)} \\ &= 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A \cap B) && \text{by (P6).} \end{aligned}$$

M8. Show that if B is an event such that $\mathbb{P}(B) = 1$, then for all events A, $\mathbb{P}(A \cap B) = \mathbb{P}(A)$. (Hint: your answer *should not* assume that $B = \Omega$! It's certainly possible to have $\mathbb{P}(B) = 1$ for a set $B \neq \Omega$.)

Answer

Since $A \cup B \supseteq B$ for any event A, it holds that $\mathbb{P}(A \cup B) \geq \mathbb{P}(B) = 1$. Hence, $\mathbb{P}(A \cup B) = 1$ by axiom (P1) and

$$\begin{aligned} \mathbb{P}(A \cap B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) \text{ by property (P6)} \\ &= \mathbb{P}(A) + 1 - 1 = \mathbb{P}(A) \,. \end{aligned}$$

M9. Consider events A, B, and C, which can occur in some experiment. Show that the probability that only A occurs (and not B or C) is equal to $\mathbb{P}(A \cup B \cup C) - \mathbb{P}(B) - \mathbb{P}(C) + \mathbb{P}(B \cap C)$.

Answer

The event "only A occurs and not B or C" is the event $A \cap (B \cup C)^c$. The trick in the calculation below is a common one: any event D can be split into two disjoint components by choosing an arbitrary other event A and writing

$$D = \Omega \cap D = (A \cap D) \cup (A^c \cap D) \,.$$

We use this below with the choice $D = (B \cup C)^c$:

$$\begin{aligned}\mathbb{P}(A \cap (B \cup C)^c) &= \mathbb{P}((B \cup C)^c) - \mathbb{P}(A^c \cap (B \cup C)^c) \\ &= \mathbb{P}((B \cup C)^c) - \mathbb{P}((A \cup B \cup C)^c) \text{ by De Morgan's law} \\ &= 1 - \mathbb{P}(B \cup C) - 1 + \mathbb{P}(A \cup B \cup C) \text{ by (P4).}\end{aligned}$$

We cancel the ones and use that $\mathbb{P}(B \cup C) = \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(B \cap C)$ according to (P6) to obtain the desired result.

Desserts

Still hungry for more? Try these if you want to push yourself further. (These are mostly harder than I'd expect you to answer in an exam, or involve non-examinable material.)

D1. Show that for $n > 0$,

$$\sum_{i=0}^{n} (-1)^i \binom{n}{i} = 0\,.$$

Answer

We know from the binomial theorem that for $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}\,.$$

Substituting $a = -1$ and $b = 1$ gives the required result.

D2. The famous **Birthday Problem**. If there are n people in a room, what is the probability that at least two people have a common birthday? (Ignore leap years for simplicity!)

Answer

As with many problems of this sort, it is easier to calculate the probability of the complementary event: that is, the probability that everybody has a distinct birthday. With n people and 365 days in the year, there is a total of 365^n possible outcomes, $|\Omega| = 365^n$. All outcomes have equal probability. How many of these have all n birthdays distinct? There are

$$365 \cdot 364 \cdot \cdots \cdot (365 - n + 1) = \frac{365!}{(365-n)!}$$

outcomes of this kind because we have 365 possible choices for the first person, then 364 for the second, and so on, and $365 - (n-1)$ choices for the n-th person. Thus the probability that at least two people have a common birthday is given by

$$1 - \frac{365!}{(365-n)!}\frac{1}{365^n}\,.$$

D3. Let k be a non-negative integer. How many distinct integer-valued vectors $(n_1, n_2, \ldots, n_r)$ are there which satisfy both of the following constraints?

i) $n_j \geq k$ for all $j = 1, 2, \ldots, r$
ii) $n_1 + n_2 + \cdots + n_r = n$.

Answer

Let $m_j = n_j - (k-1) \geq 1$. The number of solutions of

$$n_1 + n_2 + \cdots + n_r = n, \quad \text{with } n_j \geq k \text{ for all } j$$

is equal to the number of solutions to

$$m_1 + m_2 + \cdots + m_r = n - r(k-1), \quad \text{with } m_j \geq 1 \text{ for all } j.$$

The number of such solutions is given by

$$C_{r-1}^{n-r(k-1)-1} = \binom{n-r(k-1)-1}{r-1}.$$

Alternatively, this problem is equivalent to having to divide n sweets among r children, such that each child gets at least k sweets each. We can do this by first of all giving each child k sweets, meaning that there are now $n - rk$ sweets left to divide (where we are allowed to give some children zero extra sweets). The number of ways of doing this is

$$C_{r-1}^{(n-rk)+r-1} = \binom{n-r(k-1)-1}{r-1}.$$

D4. Calculating combinations for large integers can get a bit messy, but **Stirling's Approximation** can sometimes be very useful: this says that as n gets large, $n!$ can be well-approximated by

$$\sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

Use Stirling's approximation to show that, for large n, the number of ways of choosing n out of $2n$ distinct items is approximately

$$\frac{4^n}{\sqrt{\pi n}}.$$

Answer

The number of ways of choosing n out of $2n$ items is $\binom{2n}{n} = (2n)!/(n!n!)$. Replacing the factorials by Stirling's approximation shows that for large n, this number is well-approximated as follows:

$$\binom{2n}{n} = \frac{(2n)!}{n!n!} \approx \frac{\sqrt{2\pi 2n}\left(\frac{2n}{e}\right)^{2n}}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n \sqrt{2\pi n}\left(\frac{n}{e}\right)^n} = \frac{4^n}{\sqrt{\pi n}}.$$